Exponential XOR hardness amplification through random simplification

William M. Hoza

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Allowed standard prerequisites (editorial).

- Decision trees, conditional expectation and product distributions. Author Definitions 2.1, 2.4, 2.5; necessary definitions and full author Claim3.3 retained.

Definitions and notation (author Section 2)

We write $\mathbb{N}$ to denote the set of non-negative integers. In the introduction, we worked with functions $f : \{0,1\}^n \to \{0,1\}$. Going forward, it will be more convenient to encode a bit $b \in \{0,1\}$ as the value $(-1)^b$. Thus, we will work with functions $f : \{\pm 1\}^n \to \{\pm 1\}$. We continue to use the notation $f^{\oplus t}$, but now $f^{\oplus t}$ denotes the product of t copies of f on independent inputs. We use the following notation to describe decision trees.

Definition 2.1 (Decision trees). For a function $f : \{\pm 1\}^n \to \{\pm 1\}$, we define $\mathrm{DTDepth}(f)$ to be the minimum depth of a decision tree computing f. In the other direction, for a parameter $D \in \mathbb{N}$, we define $\mathrm{DTDepth}[D]$ to be the class of all functions $f : \{\pm 1\}^n \to \{\pm 1\}$ that can be computed by depth-D decision trees. (The parameter n will always be clear from context.) We denote random variables using boldface. We write $\mathbf{x} \sim \mu$ to indicate that the random variable $\mathbf{x}$ is sampled from the distribution μ.

Definition 2.4 (Tensor product of probability distributions). Let $\mu_1, \dots, \mu_t$ be probability distributions over the spaces $\Omega_1, \dots, \Omega_t$. Sample $\mathbf{x}^1 \sim \mu_1, \dots, \mathbf{x}^t \sim \mu_t$ independently. The tensor product $\mu_1 \otimes \cdots \otimes \mu_t$ is the probability distribution of $(\mathbf{x}^1, \dots, \mathbf{x}^t)$. As a special case, we define

$$\mu^{\otimes t} = \underbrace{\mu \otimes \mu \otimes \cdots \otimes \mu}_{t \text{ copies}}.$$

We use the following standard definition to reason about average-case hardness of $\{\pm 1\}$-valued functions.

Definition 2.5 (Correlation). Let $g, h : \{\pm 1\}^n \to \mathbb{R}$ be functions and let μ be a distribution over $\{\pm 1\}^n$. We define

$$\mathrm{Corr}_\mu(g, h) = \mathbb{E}_{\mathbf{x}\sim\mu}[g(\mathbf{x}) \cdot h(\mathbf{x})].$$

More generally, if $\mathcal{C}$ is a class of functions $g : \{\pm 1\}^n \to \mathbb{R}$, then we define

$$\mathrm{Corr}_\mu(\mathcal{C}, h) = \max_{g\in\mathcal{C}} \mathrm{Corr}_\mu(g, h).$$

If μ is omitted, then by default it is assumed to be the uniform distribution over $\{\pm 1\}^n$. If g and h are $\{\pm 1\}$-valued, then a bound $|\mathrm{Corr}(g, h)| \le \varepsilon$ is equivalent to the statement that g agrees with h on at most a $(1/2 + \varepsilon/2)$-fraction of inputs, because for any two $\{0, 1\}$-valued random variables $\mathbf{a}, \mathbf{b}$, we have $\Pr[\mathbf{a} = \mathbf{b}] = \frac{1}{2} + \frac{1}{2}\mathbb{E}[(-1)^{\mathbf{a}} \cdot (-1)^{\mathbf{b}}]$.

Definition 2.9 (Log-concave). A function $f : [0, \infty) \to (0, \infty)$ is log-concave if $\log f$ is concave, i.e., for every $x, y \in [0, \infty)$ and $\lambda \in (0, 1)$, we have $f(x)^\lambda \cdot f(y)^{1-\lambda} \le f(\lambda x + (1-\lambda)y)$. If f is log-concave, then by induction on t, we have $\prod_{i=1}^t f(x_i) \le f(\bar{x})^t$ where $\bar{x} = \frac{1}{t}\sum_{i=1}^t x_i$.

Definition 2.2 (Max-divergence). Let μ and $\widetilde{\mu}$ be discrete probability distributions over some set Ω. The max-divergence of $\widetilde{\mu}$ from μ is defined by

$$D_\infty(\widetilde{\mu}\|\mu) = \ln\left(\max_{x\in\Omega}\left(\frac{\Pr[\widetilde{\mathbf{x}} = x]}{\Pr[\mathbf{x} = x]}\right)\right),$$

where $\mathbf{x} \sim \mu$ and $\widetilde{\mathbf{x}} \sim \widetilde{\mu}$. Max-divergence and total variation distance are related by the following lemma.

Lemma 2.3 (Low max-divergence ⇒ low total variation distance). Let μ and $\widetilde{\mu}$ be discrete probability distributions over the same set Ω. Let $\varepsilon = D_\infty(\widetilde{\mu}\|\mu)$. There exists a probability distribution μ' such that μ can be written as a convex combination $\mu = (1-\varepsilon)\cdot\widetilde{\mu} + \varepsilon \cdot \mu'$. Moreover, the total variation distance between μ and $\widetilde{\mu}$ is at most ε.

Proof. If $\varepsilon = 0$, the lemma is trivial, so assume $\varepsilon > 0$. For each $x \in \Omega$, define

$$p(x) = \frac{\Pr[\mathbf{x} = x] - (1-\varepsilon)\Pr[\widetilde{\mathbf{x}} = x]}{\varepsilon},$$

where $\mathbf{x} \sim \mu$ and $\widetilde{\mathbf{x}} \sim \widetilde{\mu}$. Then $\sum_{x\in\Omega} p(x) = 1$. Furthermore, $p(x) \ge 0$, because

$$(1-\varepsilon)\Pr[\widetilde{\mathbf{x}} = x] \le (1-\varepsilon)\cdot e^\varepsilon \cdot \Pr[\mathbf{x} = x] \le \Pr[\mathbf{x} = x].$$

Therefore, $p(\cdot)$ is a probability mass function, and we can let μ' be the corresponding probability distribution. For the "moreover" part, observe that for any $S \subseteq \Omega$, we have

$$\Pr[\mathbf{x} \in S] = (1-\varepsilon)\cdot\Pr[\widetilde{\mathbf{x}} \in S] + \varepsilon \cdot \Pr[\mathbf{x}' \in S],$$

where $\mathbf{x}' \sim \mu'$. Therefore,

$$\Pr[\mathbf{x} \in S] \le \Pr[\widetilde{\mathbf{x}} \in S] + \varepsilon \cdot \Pr[\mathbf{x}' \in S] \le \Pr[\widetilde{\mathbf{x}} \in S] + \varepsilon,$$

and

$$\Pr[\mathbf{x} \in S] \ge (1-\varepsilon)\cdot\Pr[\widetilde{\mathbf{x}} \in S] \ge \Pr[\widetilde{\mathbf{x}} \in S] - \varepsilon.$$

□

Definition 2.6 (Generalized restriction). A generalized restriction is a function $\pi : \{\pm 1\}^r \to \{\pm 1\}^n$. If $f : \{\pm 1\}^n \to \{\pm 1\}$ is a Boolean function, then we define $g|_\pi$ to be the composition $g \circ \pi$. That is, $g|_\pi : \{\pm 1\}^r \to \{\pm 1\}$ is given by $g|_\pi(x) = g(\pi(x))$. Traditional restrictions can be viewed as a special case of generalized restrictions as follows.

Definition 2.7 (Traditional restrictions as generalized restrictions). A restriction is a string $\rho \in \{+1, -1, \star\}^n$. We identify ρ with a generalized restriction $\pi : \{\pm 1\}^r \to \{\pm 1\}^n$, where $r = |\rho^{-1}(\star)|$, as follows. Given $y \in \{\pm 1\}^r$, we let $\pi(y)$ be ρ, except that the i-th star is replaced with y_i for every $i \in [r]$. Next, we consider distributions over generalized restrictions, and we explain how to interpret the tensor product of such distributions.

Definition 2.8 (Tensor product of generalized restriction distributions). Let $r, n \in \mathbb{N}$, and let $\mathcal{R}$ be a distribution over generalized restrictions $\pi : \{\pm 1\}^r \to \{\pm 1\}^n$. Let $\boldsymbol{\pi}_1, \ldots, \boldsymbol{\pi}_t$ be independent samples from $\mathcal{R}$, and define $\vec{\boldsymbol{\pi}} : \{\pm 1\}^{rt} \to \{\pm 1\}^{nt}$ by concatenating, i.e.,

$$\vec{\boldsymbol{\pi}}(y^{(1)}, \ldots, y^{(t)}) = (\boldsymbol{\pi}_1(y^{(1)}), \ldots, \boldsymbol{\pi}_t(y^{(t)})).$$

Then the tensor product $\mathcal{R}^{\otimes t}$ is the distribution of the random variable $\vec{\boldsymbol{\pi}}$.

Lemma 4.2 (Tighter XOR lemma for the random simplification method). Let $n, t \in \mathbb{N}$ and let $h : \{\pm 1\}^n \to \{\pm 1\}$ be a Boolean function. Let $\mathcal{C}$ be a class of Boolean functions $g : \{\pm 1\}^{nt} \to \{\pm 1\}$ that is closed under restrictions.[1] Let $r \in \mathbb{N}$, let $\mathcal{R}$ be a distribution over generalized restrictions $\pi : \{\pm 1\}^r \to \{\pm 1\}^n$, and let μ be a distribution over $\{\pm 1\}^r$. Let $\varepsilon > 0$, and assume the following.

1. (The distribution μ approximately completes $\mathcal{R}$ to the uniform distribution.) If we sample $\boldsymbol{\pi} \sim \mathcal{R}$ and $\mathbf{y} \sim \mu$ independently, and we sample $\mathbf{x} \in \{\pm 1\}^n$ uniformly at random, then $D_\infty(\boldsymbol{\pi}(\mathbf{y}) \| \mathbf{x}) \leq \varepsilon$.

2. (The class $\mathcal{C}$ simplifies under $\mathcal{R}^{\otimes t}$.) For every $g \in \mathcal{C}$ and every $D \in \mathbb{N}$, we have
$$\Pr_{\vec{\boldsymbol{\pi}} \sim \mathcal{R}^{\otimes t}}[\mathrm{DTDepth}(g|_{\vec{\boldsymbol{\pi}}}) \geq D] \leq 2^{t-D}.$$

3. (The function h retains structure under $\mathcal{R}$.) For every $D \in \mathbb{N}$ and every $\pi \in \mathrm{Supp}(\mathcal{R})$, we have
$$\mathrm{Corr}_\mu(h|_\pi, \mathrm{DTDepth}[D]) \leq \varepsilon \cdot 2^{D/3}.$$

Then $\mathrm{Corr}(\mathcal{C}, h^{\oplus t}) \leq O(\varepsilon)^t$.

[1] Every function in $\mathcal{C}$ has domain $\{\pm 1\}^{nt}$. When we say that $\mathcal{C}$ is closed under restrictions, we are thinking of a restriction of $g \in \mathcal{C}$ as another function on nt bits that ignores some of its input variables.

Lemma 3.2 (XOR lemma for decision trees under robust hardness assumptions, general version). Let $h_1, \ldots, h_t : \{\pm 1\}^r \to \{\pm 1\}$ be functions, and define $h(y^{(1)}, \ldots, y^{(t)}) = \prod_{i=1}^{t} h_i(y^{(i)})$. Let $\mu_1, \ldots, \mu_t$ be distributions over $\{\pm 1\}^r$, and define $\mu = \mu_1 \otimes \cdots \otimes \mu_t$. Let $\varepsilon : [0, \infty) \to (0, \infty)$ be a log-concave function, and assume that for every $i \in [t]$ and every $D \in \mathbb{N}$, we have

$$\mathrm{Corr}_{\mu_i}(h_i, \mathrm{DTDepth}[D]) \leq \varepsilon(D).$$

Then for every $D \in \mathbb{N}$, we have

$$\mathrm{Corr}_{\mu}(h, \mathrm{DTDepth}[Dt/2]) \leq O(\varepsilon(D))^t.$$

The first step of the proof of Lemma 3.2 is the following claim, which enables us to relate the success probability of a tree to the success probabilities of its subtrees.

Claim 3.3 (Law of total correlation). Let $h, T, E : \{\pm 1\}^r \to \{\pm 1\}$. Let μ be a distribution over $\{0,1\}^r$. For each $b \in \{\pm 1\}$, let $p_b = \Pr_{\mathbf{y} \sim \mu}[E(\mathbf{y}) = b]$, and let μ_b be the conditional distribution $(\mathbf{y} \sim \mu \mid E(\mathbf{y}) = b)$. Suppose that T can be decomposed in the form

$$T(y) = \begin{cases} T_{+1}(y) & \text{if } E(y) = +1, \\ T_{-1}(y) & \text{if } E(y) = -1 \end{cases}$$

for some $T_{+1}, T_{-1} : \{\pm 1\}^r \to \{\pm 1\}$. Then

$$\mathrm{Corr}_{\mu}(h, T) = \sum_{b \in \{\pm 1\}} p_b \cdot \mathrm{Corr}_{\mu_b}(h, T_b).$$

Proof.

$$\begin{aligned} \mathrm{Corr}_{\mu}(h, T) &= \mathbb{E}_{\mathbf{y} \sim \mu}[h(\mathbf{y}) \cdot T(\mathbf{y})] \\ &= \sum_{b \in \{\pm 1\}} p_b \cdot \mathbb{E}_{\mathbf{y} \sim \mu}[h(\mathbf{y}) \cdot T(\mathbf{y}) \mid E(\mathbf{y}) = b] \quad \text{(Law of total expectation)} \\ &= \sum_{b \in \{\pm 1\}} p_b \mathbb{E}_{\mathbf{y} \sim \mu_b}[h(\mathbf{y}) \cdot T_b(\mathbf{y})]. \end{aligned}$$

□

Next, we consider the following notion of "fair" decision trees due to Shaltiel [61].

Definition 3.4 ($(D_1, \ldots, D_t)$-fair decision trees [61]). Let $T : \{\pm 1\}^{rt} \to \{\pm 1\}$ be a decision tree and let $D_1, \ldots, D_t \in \mathbb{N}$. We say that T is $(D_1, \ldots, D_t)$-fair if for every input $\vec{y} = (y^{(1)}, \ldots, y^{(t)}) \in (\{\pm 1\}^r)^t$, for every $i \in [t]$, the computation $T(\vec{y})$ makes at most D_i queries to $y^{(i)}$. The key to proving Lemma 3.2 is to generalize Definition 3.4 to the case of a set of tuples $(D_1, \ldots, D_t)$.

Definition 3.5 (Q-fair decision trees). Let $T : \{\pm 1\}^{rt} \to \{\pm 1\}$ be a decision tree and let $Q \subseteq \mathbb{N}^t$. We say that T is Q-fair if for every input $\vec{y} = (y^{(1)}, \ldots, y^{(t)}) \in (\{\pm 1\}^r)^t$, there is some tuple $(D_1, \ldots, D_t) \in Q$ such that for every $i \in [t]$, the computation $T(\vec{y})$ makes at most D_i queries to $y^{(i)}$. We emphasize that the tuple $(D_1, \ldots, D_t)$ is permitted to vary from one input $\vec{y}$ to another. Therefore, the fact that a tree is Q-fair does not necessarily imply that there is some $(D_1, \ldots, D_t) \in Q$ such that the tree is $(D_1, \ldots, D_t)$-fair. Given the concept of Q-fairness, it is relatively straightforward to prove the following claim by induction on the depth of T. The claim generalizes the analysis by Shaltiel [61], who considered the case of $(D_1, \ldots, D_t)$-fair decision trees and focused on the uniform distribution.

Claim 3.6 (XOR lemma for Q-fair decision trees). Let $h_1, \ldots, h_t : \{\pm 1\}^r \to \{\pm 1\}$ be functions, and define $h(y^{(1)}, \ldots, y^{(t)}) = \prod_{i=1}^t h_i(y^{(i)})$. Let $\mu_1, \ldots, \mu_t$ be distributions over $\{\pm 1\}^r$, and define $\mu = \mu_1 \otimes \cdots \otimes \mu_t$. Let $Q \subseteq \mathbb{N}^t$ and let $T : \{\pm 1\}^{rt} \to \{\pm 1\}$ be a Q-fair decision tree. Then

$$\operatorname{Corr}_\mu(h, T) \leq \sum_{(D_1,\ldots,D_t) \in Q} \prod_{i=1}^t \operatorname{Corr}_{\mu_i}(h_i, \operatorname{DTDepth}[D_i]).$$

Proof. Assume without loss of generality that T never queries the same variable twice. For the base case, if T has depth 0, then T is a constant function, so

$$|\operatorname{Corr}_\mu(h, T)| = \prod_{i=1}^t \left| \mathbb{E}_{\mathbf{y}^{(i)} \sim \mu_i}[h_i(\mathbf{y}^{(i)})] \right| = \prod_{i=1}^t \operatorname{Corr}_{\mu_i}(h_i, \operatorname{DTDepth}[0]).$$

Since T is Q-fair, Q must be nonempty. The lemma follows because $\operatorname{Corr}_{\mu_i}(h_i, \operatorname{DTDepth}[0]) \leq \operatorname{Corr}_{\mu_i}(h_i, \operatorname{DTDepth}[D_i])$ for every $D_i \in \mathbb{N}$. For the inductive step, let $y_{j^*}^{(i^*)}$ be the variable queried by the root of the tree. Let T_{+1} and T_{-1} be the children of the root, corresponding to the cases $y_{j^*}^{(i^*)} = +1$ and $y_{j^*}^{(i^*)} = -1$ respectively. Define

$$Q' = \{(D_1, \ldots, D_{i^*-1}, D_{i^*} - 1, D_{i^*+1}, \ldots, D_t) : (D_1, \ldots, D_t) \in Q \text{ and } D_{i^*} \neq 0\}.$$

Then T_{+1} and T_{-1} are both Q'-fair. For each $b \in \{\pm 1\}$, define

$$p_b = \Pr_{\mathbf{y}^{(i^*)} \sim \mu_{i^*}}[\mathbf{y}_{j^*}^{(i^*)} = b].$$

Let $\mu_{i^*}^b$ be the conditional distribution $(\mathbf{y}^{(i^*)} \sim \mu_{i^*} \mid \mathbf{y}_{j^*}^{(i^*)} = b)$, and for $i \neq i^*$, let $\mu_i^b = \mu_i$. Let $\mu^b = \mu_1^b \otimes \cdots \otimes \mu_t^b$. By Claim 3.3 and the induction hypothesis, we have

$$\begin{aligned}
\operatorname{Corr}_\mu(h, T) &= \sum_{b \in \{\pm 1\}} p_b \cdot \operatorname{Corr}_{\mu^b}(h, T_b) \\
&\leq \sum_{b \in \{\pm 1\}} p_b \cdot \sum_{(D_1,\ldots,D_t) \in Q'} \prod_{i=1}^t \operatorname{Corr}_{\mu_i^b}(h_i, \operatorname{DTDepth}[D_i]) \\
&= \sum_{(D_1,\ldots,D_t) \in Q'} \left(\sum_{b \in \{\pm 1\}} p_b \cdot \operatorname{Corr}_{\mu_{i^*}^b}(h_{i^*}, \operatorname{DTDepth}[D_{i^*}]) \right) \cdot \Pi_{\neq i^*}(D_1, \ldots, D_t)
\end{aligned}$$

where $\Pi_{\neq i^*}(D_1,\dots,D_t) = \prod_{i\in[t],i\neq i^*} \mathrm{Corr}_{\mu_i}(h_i, \mathrm{DTDepth}[D_i])$. Now we bound the inner sum. By Claim 3.3, for any D_{i^*}, we have

$$\mathrm{Corr}_{\mu_{i^*}}(h_{i^*}, \mathrm{DTDepth}[D_{i^*}+1]) \geq \sum_{b\in\{\pm1\}} p_b \cdot \mathrm{Corr}_{\mu^b_{i^*}}(h_{i^*}, \mathrm{DTDepth}[D_{i^*}]),$$

because we can approximate h_{i^*} with respect to μ_{i^*} by first querying $y^{(i^*)}_{j^*}$ and then using optimal subtrees of depth D_{i^*}. For every $(D_1,\dots,D_t)\in Q'$, we have $(D_1,\dots,D_{i^*-1},D_{i^*}+1,D_{i^*+1},\dots,D_t)\in Q$. Therefore,

$$\mathrm{Corr}_{\mu}(h,T) \leq \sum_{(D_1,\dots,D_t)\in Q} \prod_{i=1}^{t} \mathrm{Corr}_{\mu_i}(h_i, \mathrm{DTDepth}[D_i]).$$

□

Given Claim 3.6, our XOR lemma for decision trees under a robust hardness assumption (Lemma 3.2) readily follows, as we now show.

Proof of Lemma 3.2. Let $T:\{\pm1\}^{rt}\to\{\pm1\}$ be a decision tree of depth at most $Dt/2$. Let Q be the set of t-tuples $(D_1,\dots,D_t)\in\mathbb{N}^t$ such that (1) $D_1+\cdots+D_t \leq Dt$ and (2) D_i is an integer multiple of $\lceil D/2\rceil$ for every i. We claim that T is Q-fair. Indeed, let $\vec{y} = (y^{(1)},\dots,y^{(t)})$ be any input, and let D_i be the number of queries that $T(\vec{y})$ makes to $y^{(i)}$. Let D'_i be the smallest integer multiple of $\lceil D/2\rceil$ such that $D_i \leq D'_i$. Then $D'_i \leq D_i + (\lceil D/2\rceil - 1)$, and hence $D'_1+\cdots+D'_t \leq Dt/2 + t\cdot(\lceil D/2\rceil - 1) \leq Dt$, showing that $(D'_1,\dots,D'_t)\in Q$. Therefore, by Claim 3.6,

$$\mathrm{Corr}_{\mu}(h,T) \leq \sum_{(D_1,\dots,D_t)\in Q} \prod_{i=1}^{t} \mathrm{Corr}_{\mu_i}(h_i, \mathrm{DTDepth}[D_i]).$$

For any $(D_1,\dots,D_t)\in Q$, we can define $(D'_1,\dots,D'_t)$ such that $D'_i \geq D_i$ and $D'_1+\cdots+D'_t$ is exactly Dt rather than being at most Dt. Then $\mathrm{Corr}_{\mu_i}(h_i, \mathrm{DTDepth}[D_i]) \leq \mathrm{Corr}_{\mu_i}(h_i, \mathrm{DTDepth}[D'_i])$, so

$$\begin{aligned}
\mathrm{Corr}_{\mu}(h,T) &\leq \sum_{(D_1,\dots,D_t)\in Q} \prod_{i=1}^{t} \mathrm{Corr}_{\mu_i}(h_i, \mathrm{DTDepth}[D'_i]) \\
&\leq \sum_{(D_1,\dots,D_t)\in Q} \prod_{i=1}^{t} \varepsilon(D'_i) \\
&\leq \sum_{(D_1,\dots,D_t)\in Q} \varepsilon(D)^t && \text{(Log-concavity)} \\
&= |Q|\cdot\varepsilon(D)^t.
\end{aligned}$$

To bound $|Q|$, observe that if $(D_1,\dots,D_t)\in Q$, then we can write $D_i = c_i\cdot\lceil D/2\rceil$ for some nonnegative integers $c_1,\dots,c_t$. Furthermore, $Dt \geq \sum_i c_i\cdot\lceil D/2\rceil \geq (D/2)\cdot\sum_i c_i$, so $c_1+\cdots+c_t \leq 2t$. Therefore, $|Q|$ is at most the number of ways that $2t$ can be partitioned into $t+1$ nonnegative integers, which is precisely $\binom{3t}{t}$. Thus,

$$\mathrm{Corr}_{\mu}(h,T) \leq \binom{3t}{t}\cdot\varepsilon(D)^t \leq O(\varepsilon(D))^t.$$

□

Proof. Fix any $g \in \mathcal{C}$. Our job is to analyze the correlation between g and $h^{\oplus t}$ under a uniform random input. By Lemma 2.3, we can sample a uniform random input by the following procedure.

1. Sample $\vec{\boldsymbol{\pi}} = (\boldsymbol{\pi}_1, \ldots, \boldsymbol{\pi}_t) \sim \mathcal{R}^{\otimes t}$.
2. Sample $\vec{\mathbf{y}} = (\mathbf{y}^{(1)}, \ldots, \mathbf{y}^{(t)}) \sim \mu^{\otimes t}$.
3. Sample $\vec{\mathbf{e}} = (\mathbf{e}^{(1)}, \ldots, \mathbf{e}^{(t)}) \sim (\mu')^{\otimes t}$, where μ' is the distribution over $\{\pm 1\}^n$ from Lemma 2.3.
4. Sample $\mathbf{I} \subseteq [t]$ where $\Pr[i \in \mathbf{I}] = 1 - \varepsilon$ independently for every i.
5. Output the string $\vec{\mathbf{x}} = (\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(t)}) \in \{\pm 1\}^{nt}$, where

$$\mathbf{x}^{(i)} = \begin{cases} \boldsymbol{\pi}_i(\mathbf{y}^{(i)}) & \text{if } i \in \mathbf{I} \\ \mathbf{e}^{(i)} & \text{if } i \notin \mathbf{I}. \end{cases}$$

Let $\mathbf{g} : \{\pm 1\}^{nt} \to \{\pm 1\}$ be the function obtained from g by plugging $\mathbf{e}^{(i)}$ into each block $i \notin \mathbf{I}$ and leaving the blocks in $\mathbf{I}$ alive. Since $\mathbf{g}$ ignores the variables in blocks outside $\mathbf{I}$, we have

$$g(\vec{\mathbf{x}}) = \mathbf{g}|_{\vec{\boldsymbol{\pi}}}(\vec{\mathbf{y}}).$$

Similarly, define $\mathbf{h} : \{\pm 1\}^{nt} \to \{\pm 1\}$ by the formula

$$\mathbf{h}(\vec{x}) = \left(\prod_{i \in \mathbf{I}} h_i(x^{(i)})\right) \cdot \left(\prod_{i \notin \mathbf{I}} h_i(\mathbf{e}^{(i)})\right),$$

so that $h^{\oplus t}(\vec{\mathbf{x}}) = \mathbf{h}|_{\vec{\boldsymbol{\pi}}}(\vec{\mathbf{y}})$. That way,

$$\mathrm{Corr}(g, h^{\oplus t}) = \mathbb{E}[g(\vec{\mathbf{x}}) \cdot h^{\oplus t}(\vec{\mathbf{x}})] = \mathbb{E}_{\mathbf{I}, \vec{\mathbf{e}}, \vec{\boldsymbol{\pi}}}[\mathrm{Corr}_{\mu^{\otimes t}}(\mathbf{g}|_{\vec{\boldsymbol{\pi}}}, \mathbf{h}|_{\vec{\boldsymbol{\pi}}})].$$

Let $\mathbf{D} = \lfloor \mathrm{DTDepth}(\mathbf{g}|_{\vec{\boldsymbol{\pi}}})/|\mathbf{I}| \rfloor$. Then

$$\mathbb{E}_{\mathbf{I}, \vec{\mathbf{e}}, \vec{\boldsymbol{\pi}}}[\mathrm{Corr}_{\mu^{\otimes t}}(\mathbf{g}|_{\vec{\boldsymbol{\pi}}}, \mathbf{h}|_{\vec{\boldsymbol{\pi}}})] \leq \mathbb{E}_{\mathbf{I}, \vec{\mathbf{e}}, \vec{\boldsymbol{\pi}}}[\mathrm{Corr}_{\mu^{\otimes t}}(\mathbf{h}|_{\vec{\boldsymbol{\pi}}}, \mathrm{DTDepth}[(\mathbf{D}+1) \cdot |\mathbf{I}|])].$$

Let $\boldsymbol{\pi}_{\mathbf{I}} = (\boldsymbol{\pi}_i)_{i \in \mathbf{I}}$, and define $h_I : \{\pm 1\}^{n|I|} \to \{\pm 1\}$ by $h_I((x^{(i)})_{i \in I}) = \prod_{i \in I} h(x^{(i)})$. Then for any fixing of $\mathbf{I}, \vec{\mathbf{e}}, \vec{\boldsymbol{\pi}}$, we have

$$\mathrm{Corr}_{\mu^{\otimes t}}(\mathbf{h}|_{\vec{\boldsymbol{\pi}}}, \mathrm{DTDepth}[(\mathbf{D}+1) \cdot |\mathbf{I}|]) = \mathrm{Corr}_{\mu^{\otimes |\mathbf{I}|}}(h_{\mathbf{I}}|_{\boldsymbol{\pi}_{\mathbf{I}}}, \mathrm{DTDepth}[(\mathbf{D}+1) \cdot |\mathbf{I}|]).$$

Now we apply Lemma 3.2. For each $i \in \mathbf{I}$ and each $D \in \mathbb{N}$, we have

$$\mathrm{Corr}_{\mu}(h|_{\boldsymbol{\pi}_i}, \mathrm{DTDepth}[D]) \leq \varepsilon \cdot 2^{D/3}.$$

Furthermore, the function $\varepsilon(D) = \varepsilon \cdot 2^{D/3}$ is log-concave. Therefore, Lemma 3.2 guarantees that

$$\mathrm{Corr}_{\mu^{\otimes |\mathbf{I}|}}(h_{\mathbf{I}}|_{\boldsymbol{\pi}_{\mathbf{I}}}, \mathrm{DTDepth}[(\mathbf{D}+1) \cdot |\mathbf{I}|]) \leq O(\varepsilon \cdot 2^{2 \cdot (\mathbf{D}+1)/3})^{|\mathbf{I}|} = O(\varepsilon \cdot 2^{2\mathbf{D}/3})^{|\mathbf{I}|}.$$

Thus, overall, we get

$$\mathrm{Corr}(g, h^{\oplus t}) \leq \mathbb{E}_{\mathbf{I}, \vec{\mathbf{e}}, \vec{\boldsymbol{\pi}}}\left[O\left(\varepsilon \cdot 2^{2\mathbf{D}/3}\right)^{|\mathbf{I}|}\right].$$

Now consider any fixing of $\mathbf{I}$ and $\vec{\mathbf{e}}$. Since $\mathcal{C}$ is closed under restrictions, $\mathbf{g} \in \mathcal{C}$. Therefore, our simplification assumption tells us that for every $D \in \mathbb{N}$, we have

$$\Pr_{\vec{\boldsymbol{\pi}}}[\mathbf{D} = D] \leq 2^{t - D \cdot |\mathbf{I}|}.$$

Consequently,

$$\begin{aligned}
\mathbb{E}_{\vec{\boldsymbol{\pi}}}\left[O\left(\varepsilon \cdot 2^{2\mathbf{D}/3}\right)^{|\mathbf{I}|}\right] &= \sum_{D=0}^{\infty} \Pr_{\vec{\boldsymbol{\pi}}}[\mathbf{D} = D] \cdot O\left(\varepsilon \cdot 2^{2D/3}\right)^{|\mathbf{I}|} \\
&\leq 2^t \cdot \sum_{D=0}^{\infty} O\left(\varepsilon \cdot 2^{-D/3}\right)^{|\mathbf{I}|} \\
&\leq 2^t \cdot O\left(\sum_{D=0}^{\infty} \varepsilon \cdot 2^{-D/3}\right)^{|\mathbf{I}|} \\
&= 2^t \cdot O(\varepsilon)^{|\mathbf{I}|}.
\end{aligned}$$

Therefore, our overall bound is given by

$$\begin{aligned}
\mathrm{Corr}(g, h^{\oplus t}) \leq \mathbb{E}_{\mathbf{I}}[2^t \cdot O(\varepsilon)^{|\mathbf{I}|}] &= 2^t \cdot \sum_{I \subseteq [t]} \Pr[\mathbf{I} = I] \cdot O(\varepsilon)^{|I|} \\
&= 2^t \cdot \sum_{I \subseteq [t]} (1-\varepsilon)^{|I|} \cdot \varepsilon^{t-|I|} \cdot O(\varepsilon)^{|I|} \\
&\leq O(\varepsilon)^t.
\end{aligned}$$

□

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